

Rishi Parthasarathy Sivakumar

rishi.sivakumar10@gmail.com | riships@umd.edu | [LinkedIn](#) | [GitHub](#) | +1 (346) 546-9667
College Park, MD, United States

Education

Master of Science in Data Science

Aug 2026 - May 2028

University of Maryland, College Park, MD, United States

Bachelor of Science in Data Science and Engineering

Dec 2022 - Aug 2026

Indian Institute of Science Education and Research (IISER) Bhopal, India (GPA: 7.39/10)

Experience

Bachelor's Thesis Student

Jan 2026 - Apr 2026

Visual Data Computing Group (VisDom) - IISER Bhopal, India - On-site

Grade: **A**

Title: Query-Aware Token Budgeting for Efficient Late-Interaction Visual Document Retrieval

- Published this work as a **first author** in the **Research Track** at **IEEE ICDM 2026**.
- Proposed a two-stage, query-aware token selection technique to address the high storage and query-time cost of late-interaction visual document retrievers.
- Demonstrated efficient late-interaction via query-specific token selection across 10 **ViDoRe** datasets in **ColModernVBERT**; achieved **98.39%** PF1 recovery at a PF8-equivalent budget via greedy selection, while token top-k method established the **practical latency-quality frontier**.

Data Science Intern

May 2025 - Jul 2025

SpectraX Management LLC - Houston, TX, United States - On-site

- Built a machine learning based **document classifier** for bulk classification of **distorted** and **scanned** documents.
- Implemented robust **preprocessing** techniques, utilized **Tesseract** for **OCR** and Python libraries such as **OpenCV** and **Scikit-Learn**, and deployed the **backend tool**.
- Developed **adaptive thresholding** and deskewing heuristics to handle extreme scan distortions, elevating classification precision to **0.91** and accelerating bulk processing speeds by **75.4%**.

Publications

- R. P. Sivakumar, R. R. Dwivedi, and V. K. Kurmi, "Query-Aware Token Budgeting for Efficient Late-Interaction Visual Document Retrieval," 2026 **IEEE International Conference on Data Mining (ICDM), 2026. [Accepted]**
<https://arxiv.org/abs/2609.07262>

Projects

Psychometric Analysis of LLMs

Dec 2024

VisDom Lab - IISER Bhopal, India

- Reproduced baseline evaluations on the **PsychoBench** framework using **PyTorch**, **Matplotlib**, **Pandas**, and **NumPy** to assess the psychological profiles of **LLMs**.
- Extended the evaluation pipeline to benchmark additional large language models across standard psychological assessment metrics, enabling **comparative analysis** of model profiles.

Classification of Satellite Images using Deep Learning

Nov 2024

- Architected a deep learning vision pipeline using **OpenCV** and **NumPy** to classify complex satellite imagery, extracting spatial features using a custom **CNN** trained on **20,000+ Kaggle images**.
- Executed a **full-lifecycle model deployment** strategy in **PyTorch**, achieving **92.4%** classification precision on the holdout test set.

Skills

Languages: Python, C, SQL | **Libraries:** PyTorch, Scikit-Learn, Pandas, NumPy, OpenCV, Matplotlib, Seaborn

Domains: Machine Learning, Deep Learning, Computer Vision, Natural Language Processing, Transformers, LLMs, Data Structures and Algorithms, A/B Testing, Feature Engineering, Statistical Modeling, Data Visualization

Developer Tools & Infrastructure: Docker, Git, Linux, Tesseract OCR, AWS, Vast.ai, Vercel, GitHub Copilot

Leadership & Community

Co-Founder | AI Student Cell, IISER Bhopal: Organized technical workshops in deep learning for 100+ students.

Department Representative | Data Science, IISER Bhopal: Coordinated academic activities among students.